

1. A car of mass 1200 kg moves up a straight road that is inclined to the horizontal at an angle α , where $\sin \alpha = \frac{1}{15}$

The total resistance to the motion of the car from non-gravitational forces is modelled as a constant force of magnitude R newtons.

At the instant when the engine of the car is working at a rate of 32 kW and the speed of the car is 20 m s^{-1} , the acceleration of the car is 0.5 m s^{-2}

Find the value of R

(5)

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Whenever a numerical value of g is required, take $g = 9.8 \text{ m s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

Answer ALL questions. Write your answers in the spaces provided.

1. A small ball of mass 0.3 kg is released from rest from a point 3.6 m above horizontal ground. The ball falls freely under gravity, hits the ground and rebounds vertically upwards.

In the first impact with the ground, the ball receives an impulse of magnitude 4.2 N s . The ball is modelled as a particle.

- (a) Find the speed of the ball immediately after it first hits the ground.

(5)

- (b) Find the kinetic energy lost by the ball as a result of the impact with the ground.

(3)

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1.

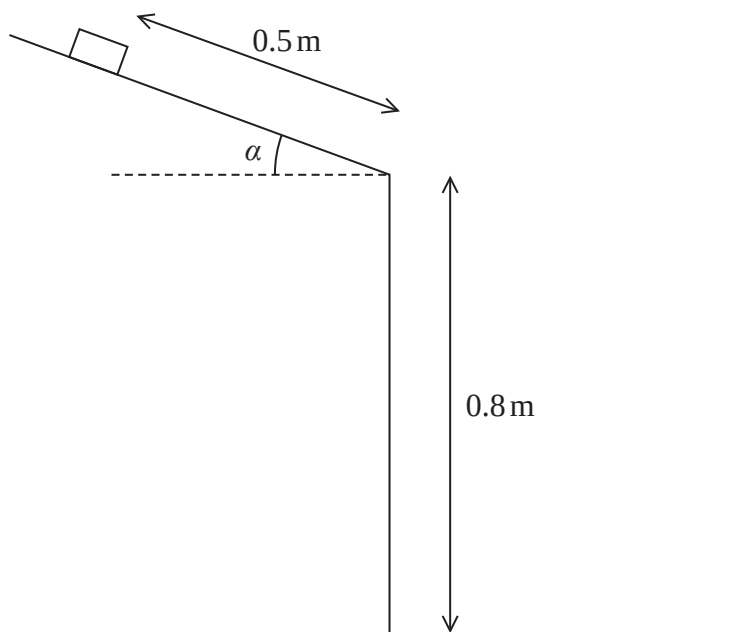


Figure 1

A small book of mass m is held on a rough straight desk lid which is inclined at an angle α to the horizontal, where $\tan \alpha = \frac{3}{4}$. The book is released from rest at a distance of 0.5 m from the edge of the desk lid, as shown in Figure 1. The book slides down the desk lid and then hits the floor that is 0.8 m below the edge of the desk lid. The coefficient of friction between the book and the desk lid is 0.4

The book is modelled as a particle which, after leaving the desk lid, is assumed to move freely under gravity.

- (a) Find, in terms of m and g , the magnitude of the normal reaction on the book as it slides down the desk lid. (2)
- (b) Use the work-energy principle to find the speed of the book as it hits the floor. (5)



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2.

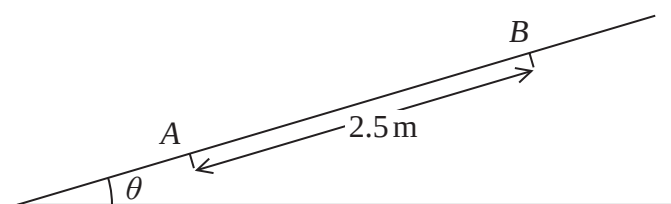


Figure 1

Figure 1 shows a ramp inclined at an angle θ to the horizontal, where $\sin \theta = \frac{2}{7}$

A parcel of mass 4 kg is projected, with speed 5 ms^{-1} , from a point A on the ramp. The parcel moves up a line of greatest slope of the ramp and first comes to instantaneous rest at the point B , where $AB = 2.5 \text{ m}$. The parcel is modelled as a particle.

The total resistance to the motion of the parcel from non-gravitational forces is modelled as a constant force of magnitude R newtons.

- (a) Use the work-energy principle to show that $R = 8.8$ (4)

After coming to instantaneous rest at B , the parcel slides back down the ramp. The total resistance to the motion of the particle is modelled as a constant force of magnitude 8.8 N .

- (b) Find the speed of the parcel at the instant it returns to A . (3)
- (c) Suggest two improvements that could be made to the model. (2)



1. A lorry of mass 16 000 kg moves along a straight horizontal road.

The lorry moves at a constant speed of 25 ms^{-1}

In an initial model for the motion of the lorry, the resistance to the motion of the lorry is modelled as having constant magnitude 16 000 N.

- (a) Show that the engine of the lorry is working at a rate of 400 kW.

(4)

The model for the motion of the lorry along the same road is now refined so that when the speed of the lorry along the same road is $V \text{ ms}^{-1}$, the resistance to the motion of the lorry is modelled as having magnitude $640V$ newtons.

Assuming that the engine of the lorry is working at the same rate of 400 kW

- (b) use the refined model to find the speed of the lorry when it is accelerating at 2.1 ms^{-2}

(6)

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2. A racing car of mass 750 kg is moving along a straight horizontal road at a constant speed of $U \text{ km h}^{-1}$. The engine of the racing car is working at a constant rate of 60 kW.

The resistance to the motion of the racing car is modelled as a force of magnitude $37.5v \text{ N}$, where $v \text{ m s}^{-1}$ is the speed of the racing car.

Using the model,

- (a) find the value of U

(4)

Later on, the racing car is accelerating up a straight road which is inclined to the horizontal at an angle α , where $\sin \alpha = \frac{5}{49}$. The engine of the racing car is working at a constant rate of 60 kW.

The total resistance to the motion of the racing car from non-gravitational forces is modelled as a force of magnitude $37.5v \text{ N}$, where $v \text{ m s}^{-1}$ is the speed of the racing car. At the instant when the acceleration of the racing car is 2 m s^{-2} , the speed of the racing car is $V \text{ m s}^{-1}$

Using the model,

- (b) find the value of V

(4)

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2. A lorry has mass 5000 kg.

In all circumstances, when the speed of the lorry is $v \text{ m s}^{-1}$, the resistance to motion of the lorry from non-gravitational forces is modelled as having magnitude $490v$ newtons.

The lorry moves along a straight horizontal road at 12 m s^{-1} , with its engine working at a constant rate of 84 kW.

Using the model,

(a) find the acceleration of the lorry.

(4)

Another straight road is inclined to the horizontal at an angle α where $\sin \alpha = \frac{1}{14}$

With its engine again working at a constant rate of 84 kW, the lorry can maintain a constant speed of $V \text{ m s}^{-1}$ up the road.

Using the model,

(b) find the value of V .

(4)

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3. A van of mass 750 kg is moving along a straight horizontal road. At the instant when the van is moving at $v \text{ m s}^{-1}$, the resistance to the motion of the van is modelled as a force of magnitude $\lambda v \text{ N}$, where λ is a constant.

The engine of the van is working at a constant rate of 18 kW.

At the instant when $v = 15$, the acceleration of the van is 0.6 m s^{-2}

- (a) Show that $\lambda = 50$

(4)

The van now moves up a straight road inclined at an angle to the horizontal, where

$$\sin \alpha = \frac{1}{15}$$

At the instant when the van is moving at $v \text{ m s}^{-1}$, the resistance to the motion of the van from non-gravitational forces is modelled as a force of magnitude $50v \text{ N}$.

When the engine of the van is working at a constant rate of 12 kW, the van is moving at a constant speed $V \text{ m s}^{-1}$

- (b) Find the value of V .

(5)

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3. A particle, P , of mass m kg is projected with speed 5 ms^{-1} down a line of greatest slope of a rough plane. The plane is inclined to the horizontal at an angle α , where $\sin \alpha = \frac{3}{5}$

The total resistance to the motion of P is a force of magnitude $\frac{1}{5} mg$

Use the work-energy principle to find the speed of P at the instant when it has moved a distance 8 m down the plane from the point of projection.

(7)

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2. A car of mass 1000 kg moves along a straight horizontal road.

In all circumstances, when the speed of the car is $v \text{ m s}^{-1}$, the resistance to the motion of the car is modelled as a force of magnitude $cv^2 \text{ N}$, where c is a constant.

The maximum power that can be developed by the engine of the car is 50 kW.

At the instant when the speed of the car is 72 km h^{-1} and the engine is working at its maximum power, the acceleration of the car is 2.25 m s^{-2}

- (a) Convert 72 km h^{-1} into m s^{-1} (1)

- (b) Find the acceleration of the car at the instant when the speed of the car is 144 km h^{-1} and the engine is working at its maximum power. (7)

The maximum speed of the car when the engine is working at its maximum power is $V \text{ km h}^{-1}$.

- (c) Find, to the nearest whole number, the value of V . (4)

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3. A plane is inclined to the horizontal at an angle α , where $\tan \alpha = \frac{3}{4}$

A particle P is held at rest at a point A on the plane.

The particle P is then projected with speed 25 m s^{-1} from A , up a line of greatest slope of the plane.

In an initial model, the plane is modelled as being smooth and air resistance is modelled as being negligible.

Using this model and the principle of conservation of mechanical energy,

- (a) find the speed of P at the instant when it has travelled a distance $\frac{25}{6} \text{ m}$ up the plane from A .

(4)

In a refined model, the plane is now modelled as being rough, with the coefficient of friction between P and the plane being $\frac{3}{5}$

Air resistance is still modelled as being negligible.

Using this refined model and the work-energy principle,

- (b) find the speed of P at the instant when it has travelled a distance $\frac{25}{6} \text{ m}$ up the plane from A .

(8)

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4. A small ball, of mass m , is thrown vertically upwards with speed $\sqrt{8gH}$ from a point O on a smooth horizontal floor. The ball moves towards a smooth horizontal ceiling that is a vertical distance H above O . The coefficient of restitution between the ball and the ceiling is $\frac{1}{2}$

In a model of the motion of the ball, it is assumed that the ball, as it moves up or down, is subject to air resistance of constant magnitude $\frac{1}{2}mg$.

Using this model,

- (a) use the work-energy principle to find, in terms of g and H , the speed of the ball immediately before it strikes the ceiling, (5)
- (b) find, in terms of g and H , the speed of the ball immediately before it strikes the floor at O for the first time. (5)

In a simplified model of the motion of the ball, it is assumed that the ball, as it moves up or down, is subject to no air resistance.

Using this simplified model,

- (c) explain, without any detailed calculation, why the speed of the ball, immediately before it strikes the floor at O for the first time, would still be less than $\sqrt{8gH}$ (1)



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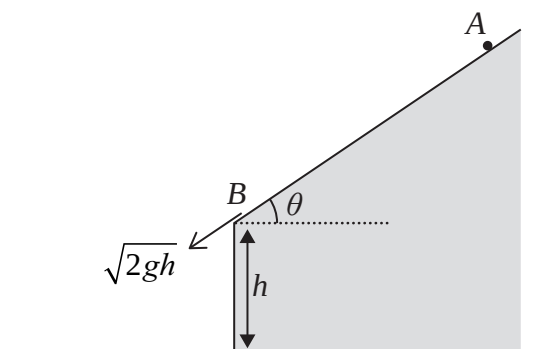


Figure 1

Figure 1 shows part of the end elevation of a building which sits on horizontal ground. The side of the building is vertical and has height h .

A small stone of mass m is at rest on the roof of the building at the point A . The stone slides from rest down a line of greatest slope of the roof and reaches the edge B of the roof with speed $\sqrt{2gh}$.

The stone then moves under gravity before hitting the ground with speed W .

In a model of the motion of the stone **from B to the ground**

- the stone is modelled as a particle
- air resistance is ignored

Using the principle of conservation of mechanical energy and the model,

(a) find W in terms of g and h .

(4)

In a model of the motion of the stone **from A to B**

- the stone is modelled as a particle of mass m
- air resistance is ignored
- the roof of the building is modelled as a rough plane inclined to the horizontal at an angle θ , where $\tan \theta = \frac{3}{4}$
- the coefficient of friction between the stone and the roof is $\frac{1}{3}$
- $AB = d$

Using this model,

(b) find, in terms of m and g , the magnitude of the frictional force acting on the stone as it slides down the roof,

(3)

(c) use the work–energy principle to find d in terms of h .

(5)



3. The total mass of a cyclist and his bicycle is 100 kg.

In all circumstances, the magnitude of the resistance to the motion of the cyclist from non-gravitational forces is modelled as being kv^2 N, where $v \text{ m s}^{-1}$ is the speed of the cyclist.

The cyclist can freewheel, without pedalling, down a slope that is inclined to the horizontal at an angle α , where $\sin \alpha = \frac{1}{35}$, at a constant speed of $V \text{ m s}^{-1}$

When he is pedalling up a slope that is inclined to the horizontal at an angle β , where $\sin \beta = \frac{1}{70}$, and he is moving at the same constant speed $V \text{ m s}^{-1}$, he is working at a constant rate of P watts.

- (a) Find P in terms of V .

(7)

If he pedals and works at a rate of $35V$ watts on a horizontal road, he moves at a constant speed of $U \text{ m s}^{-1}$

- (b) Find U in terms of V .

(4)

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